

# Day 1: Insurance Premium Prediction Report

## 1. Project Objective

The objective of the Day 1 work for the Insurance Premium Prediction project is to perform initial data exploration and preprocessing. This includes loading the dataset, understanding its structure (columns, dimensions, data types), evaluating summary statistics, and identifying missing or duplicated values. Furthermore, Day 1 involves analyzing the distribution of the target variable (Total Annual Premium in INR).

## 2. Data Exploration & Findings

- Dataset Shape: The dataset consists of 10,000 records and 22 columns.
- Missing Values: Identified missing data in Roof\_Type (100), Security\_Level (1065), Previous\_Claim\_Amount\_INR (100), and Credit\_Score (100).
- Duplicates: No duplicate rows were found in the dataset.
- Target Variable: The 'Total\_Annual\_Premium\_INR' averages at ~66,980 INR, ranging from a minimum of 5,000 INR to a maximum of 701,600 INR.

## 3. Step-by-Step Implementation

Below are the key steps executed in Python, along with screenshots documenting the outputs:

### Step 3.1: Loading the Dataset

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JupyterLab Python 3 (pykernel)

```
[2]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

pd.set_option("display.max_columns", None)

df = pd.read_csv("dataset.csv")

print("Dataset loaded successfully!")
print("Shape:", df.shape)

display(df.head())
```

Dataset loaded successfully!  
Shape: (10000, 22)

|   | Property_ID | Property_Age_Years | Property_Type | Location_Type | Property_Size_Sqft | Property_Value_INR | Construction_Type | Roof_Type | Number_of_Floors | Occupancy |
|---|-------------|--------------------|---------------|---------------|--------------------|--------------------|-------------------|-----------|------------------|-----------|
| 0 | PROP-000001 | 7                  | Apartment     | Urban         | 1033.0             | 29852000.0         | Mixed             | Concrete  | 4                | Owner O   |
| 1 | PROP-000002 | 61                 | Villa         | Urban         | 1247.0             | 22572000.0         | Steel             | Metal     | 2                | Owner O   |
| 2 | PROP-000003 | 52                 | Townhouse     | Urban         | 2080.0             | 34072000.0         | Wood              | Asphalt   | 1                | Tenant O  |
| 3 | PROP-000004 | 35                 | Apartment     | Rural         | 1619.0             | 47635000.0         | Brick             | Metal     | 4                | Owner O   |
| 4 | PROP-000005 | 34                 | Apartment     | Suburban      | 1320.0             | 2896000.0          | Concrete          | Asphalt   | 2                | MI        |

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*Figure 1: Loading dataset with pandas and checking shape.*

## Step 3.2: Inspecting Dataset Information

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[3]:

```
# Dataset information
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 10000 entries, 0 to 9999
Data columns (total 22 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Property_ID                           10000 non-null  str
1   Property_Age_Years                     10000 non-null  int64
2   Property_Type                           10000 non-null  str
3   Location_Type                           10000 non-null  str
4   Property_Size_SqFt                     10000 non-null  float64
5   Property_Value_INR                     10000 non-null  float64
6   Construction_Type                       10000 non-null  str
7   Roof_Type                              9900 non-null   str
8   Number_of_Floors                       10000 non-null  int64
9   Occupancy_Type                         10000 non-null  str
10  Security_Level                          8935 non-null   str
11  Natural_Disaster_Risk                   10000 non-null  str
12  Crime_Risk                              10000 non-null  str
13  Previous_Claims_Count                   10000 non-null  int64
14  Previous_Claim_Amount_INR               9900 non-null   float64
15  Coverage_Amount_INR                     10000 non-null  float64
16  Deductible_INR                          10000 non-null  float64
17  Policy_Term_Years                       10000 non-null  int64
18  Customer_Age                           10000 non-null  int64
19  Customer_Income_INR                     10000 non-null  float64
20  Credit_Score                            9900 non-null   float64
21  Total_Annual_Premium_INR               10000 non-null  float64
dtypes: float64(8), int64(5), str(9)
memory usage: 1.7 MB
```

[ ]:

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*Figure 2: Using df.info() to check datatypes and null counts.*

### Step 3.3: Missing Values & Duplicates

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[5]:

```
# Missing values  
print(df.isnull().sum())
```

|                           |      |
|---------------------------|------|
| Property_ID               | 0    |
| Property_Age_Years        | 0    |
| Property_Type             | 0    |
| Location_Type             | 0    |
| Property_Size_SqFt        | 0    |
| Property_Value_INR        | 0    |
| Construction_Type         | 0    |
| Roof_Type                 | 100  |
| Number_of_Floors          | 0    |
| Occupancy_Type            | 0    |
| Security_Level            | 1065 |
| Natural_Disaster_Risk     | 0    |
| Crime_Risk                | 0    |
| Previous_Claims_Count     | 0    |
| Previous_Claim_Amount_INR | 100  |
| Coverage_Amount_INR       | 0    |
| Deductible_INR            | 0    |
| Policy_Term_Years         | 0    |
| Customer_Age              | 0    |
| Customer_Income_INR       | 0    |
| Credit_Score              | 100  |
| Total_Annual_Premium_INR  | 0    |

dtype: int64

[ ]:

|

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*Figure 3: Finding missing values per column.*

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[6]:

```
# Duplicate rows  
print("Duplicates:", df.duplicated().sum())
```

Duplicates: 0

[ ]:

|



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*Figure 4: Validating that there are zero duplicate rows.*

## Step 3.4: Target Variable Analysis

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```
[15]: target = "Total_Annual_Premium_INR"
display(df[target].describe())
```

|       |               |
|-------|---------------|
| count | 10000.000000  |
| mean  | 66960.980000  |
| std   | 60368.311833  |
| min   | 5000.000000   |
| 25%   | 20900.000000  |
| 50%   | 49400.000000  |
| 75%   | 83100.000000  |
| max   | 201600.000000 |

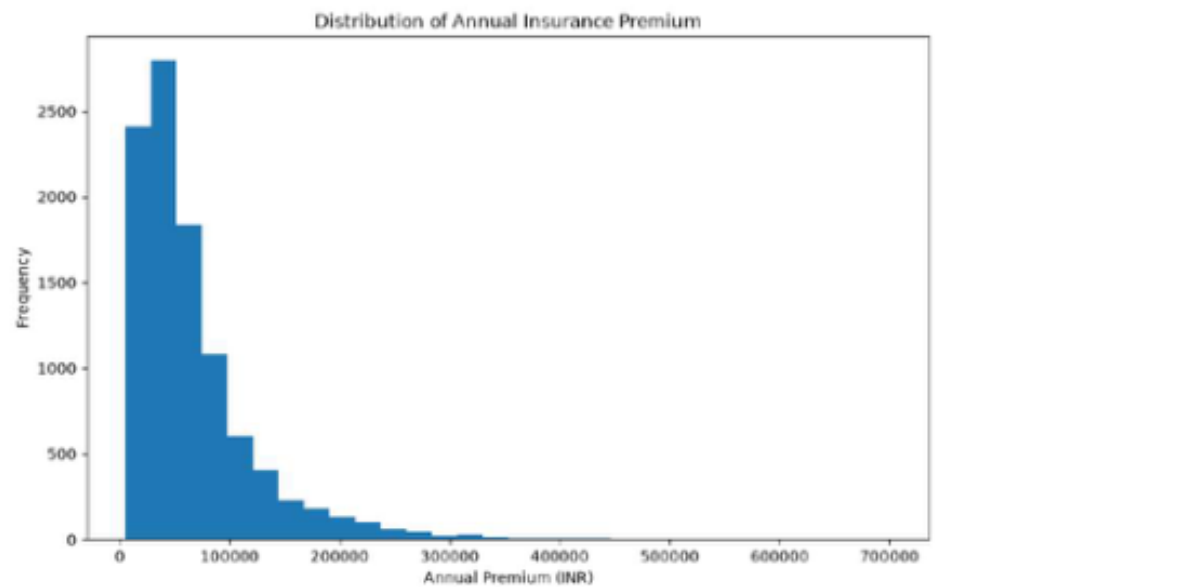
Name: Total\_Annual\_Premium\_INR, dtype: float64

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*Figure 5: Statistical summary of the target variable (Total\_Annual\_Premium\_INR).*

### Step 3.5: Visualizing Target Distribution

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*Figure 6: Histogram displaying the right-skewed distribution of insurance premiums.*

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## 4. Future Steps

Based on the Day 1 findings, the upcoming phases of the project will focus on:

1. Handling Missing Data: Imputing values for Roof\_Type, Security\_Level, Previous\_Claim\_Amount\_INR, and Credit\_Score.
2. Outlier Treatment: Addressing the right-skewness and extreme max values in the premium data.
3. Feature Engineering: Encoding categorical variables and potentially standardizing numerical ones.
4. Model Selection & Training: Building initial regression models to predict insurance premiums.